



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19 - 401 - 1

Date: 13 - 10 - 2020

EDMS 19 – 401 – 1
TECHNICAL SPECIFICATION
FOR
THREE PHASE OUTDOOR 12KV & 24 KV
VACUUM AUTOMATIC CIRCUIT RECLOSERS



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❖ توجد بنود اختيارية Option items يجب تحديدها بواسطة شركة التوزيع قبل الطرح.



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**SPECIFICATION FOR THREE PHASE OUTDOOR 12KV & 22 KV
VACUUM AUTOMATIC CIRCUIT RECLOSERS**

1. Scope

This specification covers the minimum technical requirements for design, engineering, manufacture, inspection, testing and performance of 12/24 kV three phases VACUUM Automatic Circuit Reclosers (Auto-Recloser).

Equipment should be used to isolate the faulted section(s) for 12/24 KV 50-Hz distribution feeder before the main circuit breaker which is located at the transformers station and substation (protection coordination is made between the cascaded Reclosers and the main circuit breaker). The Reclosers are out-doors installation on poles, on a distribution Overhead line circuit, and will be used for distribution automation.

2. Definitions

2.1 Automatic circuit recloser

Self-controlled device for making, carrying, and automatically interrupting and reclosing an alternating-current circuit, with a predetermined sequence of opening and reclosing followed by resetting, hold-closed, or lockout operation. It includes an assembly of control elements required to detect overcurrents and control the recloser operation.

2.2 OSM: Outdoor switch model

2.3 RTU: Remote terminal unit.

2.4 RC: Recloser control.

2.5 AR OC: Automatic Recloser overcurrent

2.6 AR SEF: Automatic Recloser sensitivity earth fault.

2.7 BF: Bolted fault protection, Very low impedance fault.



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2.8 Mechanical wear %: Corrosion percentage in circuit breaker mechanism

2.9 Contact wear%: Corrosion percentage in circuit breaker contacts.

2.10 SCADA: Supervisory control and data acquisition.

3. System Parameters

Nominal Voltage	11 kV	22 kV
Rated voltage	12kV	24 kV
System frequency	50 HZ	50 HZ
Number of phases	3	3
Method of earthing	Solidly	Solidly
System fault current(3 sec)	25 KA	20 KA

4. Service Conditions

The complete system (electrical and mechanical) must be designed to operate under the following condition:-

Annual average ambient temperature	30°C
Maximum ambient temperature	55°C
Minimum ambient temperature	- 5°C
Maximum relative humidity	95 % non-Condensing
Environmental conditions	Humid tropical climate with polluted atmosphere
Operational altitude	١٠٠٠ meter above MSL

5. Applicable codes and standards:

The equipment “Recloser” and the components supplied should be complying with the latest edition of IEC publications unless otherwise specified in this specification. The equipment/material covered in this specification should comply with the latest edition/amendment of the following codes and standards. Where any provision of this specification differs from the standards listed below, the provisions of this specification should apply.



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S.No	Standard No	Description
1.	ANSI / IEEE C 37.60 (2012) / IEC62271-111-201٢	Automatic Circuit Recloses and Fault Interrupters for AC Systems.
2.	IEC-62271-100	High-voltage alternating-current circuit-breaker.
3.	IEC-60060-2	High-voltage test techniques.
4.	IEC 60815	Guide for the selection of insulators in respect of polluted conditions.
5.	IEC 60507	Artificial pollution tests on high-voltage ceramic and glass insulators to be used on A.C. systems
6.	IEC 60044-8	Instrument transformers –Part 8:Electronic current transformers
7.	IEEE C37.235-2007	Guide for the Application of Rogowski Coils Used for Protective Relaying Purposes
8.	IEC 60255-8	Electrical relays.
9.	IEC61869-1,3	Voltage transformers
10.	EDMS 17-401-2	Technical specification for M.V voltage transformer
11.	IEC 61850	Communication protocol networks and systems for power utility automation
12.	IEC 60870-5-104	Tele-control equipment and systems.
13.	IEC60529	Classification of degrees of protection provided by enclosures.
14.	IEC 60270	High voltage test techniques-partial discharge measurements
15.	IEC-60437	Radio Interference Test on High Voltage Insulators.

6. Technical Requirements

6.1 General Data

Rated voltage KV	12KV	24KV
Rated short time current withstand, KA (3 second)	16KA	12.5KA
Breaking current RMS, KA	16KA	12.5KA
Making Current Peak, KA	40KA	31.5KA



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Rated current, A (Continuous current, A) at 40°C	630 A	630 A
Rated Impulse withstand voltage kV(1.2/50 μs) kV peak	95 KV	125 KV
Rated power frequency withstand voltage dry (1min.) kV	50 KV	60 KV
Rated power-frequency withstand voltage, kV, 10 s, wet	45 KV	50KV
Partial discharge decay level, kV, not less than (at 10 PC)	13.2 KV	17.2 KV
Cable charging current	10 A	25 A
Line charging current	2 A	5 A
Minimum number of sequential tripping operations	04	
Operating mechanism(3 poles)	magnetic actuator	
Insulation media	Solid	
Interrupting Media	Vacuum	
Minimum number of electrical operations at rated current (630Ampere)	$\geq 30,000$ CO cycles	
Minimum Interrupting life operations at rated breaking current	200 CO cycles	
Closing time, ms ,not more than	80	
Opening time, ms, not more than	45	
Breaking time, ms ,not more than	25	
Number of current sensors	6X Rogowski current sensors one per phase	
Number of voltage sensors	6X Capacitively coupled voltage sensors one per phase	
Contact resistance ($\mu\Omega$)/C.B	Less than 60$\mu\Omega$ at current supplied 100A or 25% of the continuous current rating of the C.B	



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6.2 General specifications for Auto-Reclosers

Basic operating parameters

Recloser tank construction	stainless steel or Aluminum alloy chemically pre-oxidized powder coated	
Thickness of Auto-Recloser tank	3	mm
Bushing type	Anti-fog Silicon rubber or hydrophobic-cycloaliphatic with Creepage distance not less than 440mm for 11 KV& 880 mm for 22 KV	
Maximum allowable duty cycle	O- 0,1s-CO-1s-CO-1s-CO-60s	
Recovery time for 4 th Open	60	sec
Control cable length	9 meters in tube screened against electrostatic and electromagnetic interference	m
Power supply cable	Not less than 9 meters	m
Earthing cables	Not less than 9 meters for OSM and 4 meters for RC	m
Degree of protection the Recloser and control cubicle	Sealed IP 65	
Expected life time of Recloser	25	year
Maintenance interval	Maintenance free	

Type of operating mechanism

Closing	Magnetic actuator	
Tripping	Magnetic actuator & opening spring	

Sensor accuracy

Rogowski current sensor	Not less than 1%	±%
Capacitive voltage sensor	Not less than 1%	±%

Uninterruptible Power Supply(UPS)



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Source of AC input voltage	Double pole 11000/220 or 22000/220 VAC outdoor dry type voltage transformer with burden 200 VA to be supplied with each Auto-Recloser.	V
Type, rated burden and application standard of auxiliary supply voltage transformer	Dry outdoor type ,epoxy resin , 200 VA burden	
Battery type	Sealed	
Battery voltage	12	V
Battery capacity	26	AH
Battery operating time	48	HR
Battery recharge time	≤12	HR
Battery replacement interval	≥10	Year
Battery Low Voltage indication	To be provided with alarm and recording	
RTU shutdown level	0.10.8	P.U
External power supply shutdown level	0.1.....0.8	P.U
Modem or radio power supply voltage	5...15, step 0.5	V
Modem or radio power supply max. continuous power output .	15	W
RTU (Remote terminal unit)		
RTU	included	
Communication protocol	It supports the following protocols. • Modbus protocol indication data • DNP3 protocol indication data • TCP/IP protocol indication data • Port RS-232 indication data	
OPERATION PARAMETERS PROTECTION SETTINGS RANGE		
Type of protection characteristics	Instantaneous, Definite, Standard inverse, very inverse extreme inverse.	



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Zone Sequence Coordination	Sequence control algorithm Allows Recloser to keep its trip sequence in step with another Recloser downstream should be provided.	
Operational pick-up current for phase over-current setting range	10 –6000	A
Operational pick-up current for earth –fault current setting range	10-1280	A
Current setting resolution	1	A
Sensitive Earth Fault protection (SEF)		
Operational pick up current range	1 – 80	A
Operational pick up current resolution	1	A
SEF operating time resolution	0.01	
Cold Load Pick up		
Cold Load multiplier range	1 – 2	
Cold Load multiplier resolution	0.1	
Cold Load time	1 – 60	minute
Cold Load time resolution	1	minute
Inrush harmonics filter	Present , based on detection of second harmonic	
Zone Sequence Coordination	present	
under voltage element (UV)		
Pick up phase voltage range	0.6– ١	PU
Pick up phase voltage resolution	0.01	PU
Tripping time range	0.1 – 100	Sec
Tripping time resolution	0.01	Sec
Bolted fault protection element (BF)		



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Pickup current range(instantaneous)	20....6000	A
Current unbalance protection element (CU)		
Pickup setting range	0.6....1	P.U
Time setting range	0.1....100	Sec
Voltage unbalance protection element (VU)		
Pickup setting range	0.05...1	P.U
Time setting range	0.1....100	Sec
High current lockout (Live line protection element) LL		
Pick up current	10 – 1280	A
Current setting resolution	1	A
Tripping time setting range	0....2	Sec
Time resolution	0.01	Sec
Auto – Reclose		
Duty cycle	O-0.1s-CO-1s-CO-1s-CO-60s-CO	
First Auto-reclose dead time	0.1 – 180	Sec
Second Auto-reclose dead time	1 – 180	Sec
Third Auto-reclose dead time	1 – 180	Sec
Reset time	1 – 180	Sec
Timing resolution	0.01	Sec
Voltage Reclosing control		
Voltage multiplier range	0.6 – 1	P.U
Voltage multiplier resolution	0.01	P.U
Under frequency protection element		
Pickup frequency	50 ±0.5	HZ
Pickup frequency resolution	0.01	HZ



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Tripping time	0.1 – 180	Sec
Recording		
Fault profile	The following data are presented in the Fault profile: • Phase currents • Residual current • Positive sequence current and voltage • Negative sequence current and voltage • Zero Sequence Reclosing Protection • Frequency • Active Protection • Position of the Recloser	
Event log		
Number of the events	1000	
Characteristics	date and time	
	event title	
	source of event	
	relevant state	
	critical parameter	
Integration period	5/10/15/30/60	minute
Number of registers	9000 covering interval from 31.25.....375 days	
Life time counters		
Type	Mechanical wear %	
	Contact wear%	



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	Total number of close and open operations.	
Fault counters		
Type	BF trips Tripping initiated by BF element	
	EF trips Tripping initiated by one of the following elements: EF1, EF2, EF3 and EFL	
	VU trips Tripping initiated by VU element	
	UV trips Tripping initiated by UV element	
	LS trips Tripping initiated by LS element	
	AR OC	
	AR SEF	
Loss of supply detection / Low battery warning	Present	
UPS monitoring	Present	
Mechanical position indicator	Contact position indicator (ON / OFF) colored with Green and Red) clearly visible from the ground level.	
Manual tripping	Manual tripping ring easily operable with a hook-stick.	
Lockout ring	A lockout ring should be available and can be operated manually by hook-stick to prohibit operation either remotely or from the control panel until ring resetting.	



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Drainage filter	Important to drain the condensation of the condensate if it is formed	
Anti-pumping duty	It is prevent to operate the auto-Recloser when pressing the start and stop button together	
Lifting tackle	Present	

7. Basic Features

7.1 General

Automatic Circuit Reclosers (Auto-Reclosers) should be of the three phase Solid type, suitable for outdoor, single pole mounted application and should be complete with operating mechanism, control module programming facility and all other components necessary for installation and operation.

The Auto-Reclosers should be of the minimum maintenance type and be tropicalized in accordance with service condition “environment “. Evidence of operational endurance in service, and design features to guarantee minimum maintenance performance should be furnished with the offer.

- Wide range of protection setting – solid media insulation – easy maintenance – light weight – easy installation.
- All data can download on site or remotely through communication interface.
- It is supplied complete with the RC control cubical which provides the protection, data logging and communication function in a single device.
- Auto-Recloser is based on discrimination of protection setting between different Recloser on the line.

7.2 Design Criteria

- The Auto-Recloser tank should have its own self-contained operating mechanism, and self-supporting control unit all housed in water and dust-proof stainless steel or aluminum chemically pre-oxidized powder coating with a degree of protection of IP65. Suitable means should be incorporated to eliminate the formation of condensation with minimum thickness 3mm, strong enough to support the vibration of the Re-closer. Generally, The tank and all appurtenances should be made of corrosion resistant material or provided with an impact and corrosion resistant finish and should also be



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suitable for storage in uncovered areas.

- ii) The source side and the load side of the Auto-Recloser should be indelibly marked at a suitable location to provide an easy identification.
- iii) Tripping should be initiated by over-current, earth fault and instantaneous relays. The type of over current and earth fault relays and their operating current\ time setting range should conform to the technical Design Requirements. The number of operations to lock out should be the same for over current and earth fault tripping. The Auto-Recloser dead times should be individually selectable for each Auto-Recloser operation.
- iv) Software to upload and download protection settings, to provide historical information (stored information) should be supplied.
- v) Auxiliary and tripping supplies should be an integral part of the Auto-Recloser design. Sealed, maintenance free rechargeable type battery having a service lifetime of a minimum of (10) year should be supplied.
- vi) 6XROGOWSKI Current Sensors and 6 Voltage Sensors should be provided with the Auto-Recloser for the purpose of fault detection and line measurements as stipulated in Technical Design and detection of direction of power flow. The control panel of the control cabinet should have facilities for measuring Current/Voltage for the purpose of data logging. Instantaneous Measurements, Automated Schemes etc. provision for remote control and monitoring should also be provided, and provide active and reactive power , energy per phase , power factor and frequency.
- vii) Auto-Recloser lockout should be clearly indicated. Manual lockout and reset of the Auto-Recloser should be obtained by the operation of a manual control lever.
- viii) Auto-Recloser operations should be recorded by a counter.
- ix) The Auto-Recloser should be provided with a manual operating lever for opening the Auto-Recloser.
- x) All nonmetal parts including insulating materials of cables should be able to withstand effects due to ultraviolet radiation.
- xi) The design of the Auto-Recloser should permit the application of suitable live-line techniques to disconnect and remove the Auto-Recloser from the line for maintenance at ground level or reconnect it to the line for the purpose of protecting the line personnel.
- xii) Surge Protection. The Auto-Recloser should have surge suppression on the controller and cables for both controller and radio. All power and signal cables must be protected from lightning and power surges.



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- xiii) The Auto-Recloser should be provided with suitable Steel Mounting Frame Lifting Tackle, Surge Arrester Mounting Bracket on the source and load side. The mounting bracket and hardware should be hot dip galvanized steel, each Auto-Recloser must be supplied with THREE or SIX (as an option) surge arrestors 10KA, 12KV (MAX) with continuous operating voltage $10KV \pm 5\%$ or surge arrestors 10KA, 24 KV (MAX) with continuous operating voltage $20KV \pm 5\%$ (leakage path 4 cm/KV at least)–and the mounting bracket should be equipped with each Auto-Recloser device for mounting six lightning arresters.
- xiv) Earthing terminals suitable to accommodate two Numbers ranged from 5mm dia to 15 mm dia earthing conductors should also be provided for bonding the Auto-Recloser tank, mounting frame and the surge arrester earth terminal to the local earthing electrodes.
- xv) The following tools should be supplied for each A/R (Auto-Recloser) :-
1. The required tools for commissioning and Earthing the A/R.
 2. Silicon grease for the A/R bushings.
 3. Bird guard for each bushing in the A/R.

7.3 Insulator

- i) Bushing/housing insulator should be of the Silicon rubber type or hydrophobic-cycloaliphatic and terminals should be of the Universal Clamp type to accommodate copper/aluminum conductors.
- ii) The Bushing insulator leakage distance should be 4 cm or more / 1 KV.

7.4 Controllers, Protection and Measurements

The controller cabinet should be made of stainless steel type (430S or 430L), thickness 2.5 mm, or aluminum chemically pre-oxidized powder coating with a degree of protection of IP65.

- i) The controller capable to operate in two modes: Locally and remotely operation.
In Locally mode, the Recloser can only be operated manually at the controller.
In Remotely mode, the Recloser can be operated by a System Operator at a remote location, or can be operated automatically when the controller detects a condition that requires the switch to operate. The controller should be microprocessor based with a display and a built-in keypad. In addition, the controller should be capable of providing



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protective functions, fault location, local access control, duty cycle monitoring, hot line tag, and SCADA information and control.

ii) Programmable type electronic control circuits should be used for operating the Auto-Reclosers. The Auto-Reclosers should be provided with adjustable tripping current /time-delay operation characteristics. A minimum of four (04) tripping operations to lockout should be provided and the number and sequence of operations to lockout should be adjustable in the field.

iii) The applicable over current tripping should be of the instantaneous, Definite , standard inverse, very inverse and extreme inverse types as per IEC60255 and the tripping current / time delay setting range should be as stipulated in Technical Design .

- Inrush harmonic filter should be present based on the harmonic filter, inrush multiplier technology is refused.
- Inrush restraint facilities should be provided. The tripping current / time delay setting range for earth fault and sensitive earth fault tripping should be as stipulated in Technical Design.
- If the fault cleared by any one of the Auto-Recloser operations and the Auto-Recloser remains closed, it should reset itself for another complete sequence of operations. The reset time should be adjustable from 1 - 180sec.

iii) Protection Functions:

Controller should be capable of the following minimum protective functions:

- Protection Group Selection (4 independent groups).
- Current unbalance protection.
- Voltage unbalances protection.
- Under voltage protection.
- Over voltage protection.
- Bolted fault protection.
- High current lockout (Live line protection).
- Sensitive Earth Fault protection (SEF).
- Earth fault protection.
- Over current protection.



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- Directional Blocking.
- Under/Over Frequency.
- Instantaneous over-current Protection.
- Negative sequence over-current protection.
- Definite Time Protection.
- Time-Current Curves, ANSI, IEC, Cooper Recloser, and minimum of 3 user defined curves.
- Cold Load Pick-up, with settable current threshold and time limit.
- Inrush-Restraint, based on 2nd Harmonic detection.
- Flexibility in coordination with other protective devices is provided by varied time-current characteristics from a choice of standard or customized curves, minimum trip values, reclosing and resetting time setting, and a selection of accessories.
- Sequence Coordination, settable to trip and lockout without automatic recloser or after one, two or three unsuccessful reclosing.
- The protection functions should have feature to assure instantaneous trip to the highly current and then lockout.
- Broken Conductor Protection/High Impedance Fault Protection.(Voltage unbalances)

iv)Measurements Functions:

There should be measurements functions and indication in the relay HMI for analogue inputs (current and voltage) connected to the relay, as per the following:

- Phase current.
- Current sequence components.
- Voltage sequence components.
- Three phase voltage phase and line voltages measurements.
- Active Power, Reactive Power and total Power.
- Frequency.
- Power factor.
- Active and Reactive energy.
- Maximum load.
- Residual current.
- Residual voltage.



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v) External Power Supply.

Auto-Recloser should be supplied with an outdoor voltage transformer 11000/220 or 22000/220 V with the burden of at least 200VA. Transformer burden should have adequate VA to power controller and following auxiliary functions:

- a. 220 VAC Outlet in control cabinet, to provide power for a laptop.
- b. GPRS modem system (external load supply).
- c. DC charger, which charges the battery system.
- d. Powering the controller which controls the Auto-Recloser.
- e. 4-G Gateway
- f. Battery Monitoring System (Deep Discharge Protection & Over Voltage Protection & Temperature Compensated Float Charging & Over Temperature Stop Charging)

vi) Communications.

- The controller should be complete with a built-in RTU and should have remote communications capabilities for SCADA.
- Each Auto-Recloser should be provided by 4-G Gateway

The 4-G Gateway specification is as follows:-

- 4G - gateway (class B) or (class A) with dual Sim-Card (two different vendors).
- The whole 4G - gateway Communication system should be completed and the master device should be managed to communicate with Auto-Recloser devices all time (always online).

7.5 Programming and Remote Control(SCADA)

- For remote control of the Auto-Recloser (SCADA) a microprocessor based controller with an integrated RTU should be provided, and the communication protocol should conform to the following protocols:
 - Modbus protocol indication data
 - DNP3 protocol indication data
 - TCP/IP protocol indication data
 - Port RS-232 and RS-485 indication data



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- The communication protocol should conform to a SCADA system to IEC 60870-5-104 and the updated protocols. The RTU should allow a SCADA system in ...EDC to supervise all the analogue, digital alarms and control points remotely. The interface between RTU and SCADA should be via a radio modem and 3/4-G GATEWAY.
- The interface between RTU and GATEWAY via RS232, RS485, Ethernet, etc.
- The Auto-Recloser should be supplied with only one software suitable for control and programming of the Auto-Recloser locally and remotely via Auto-Recloser software in a separate channel (not through the SCADA channel).
- The software should be able to change Auto-Recloser settings remotely through 4-G Gateway and locally through RS-232, RS-485, USB and Ethernet.
- This software should scan the status of each Auto-Recloser every certain programmed interval (scan time).
- In case of SCADA Server shutdown, The A/R controller should have a memory or a register to save faults with its times and send it again to the server in the next scan time.
- In case of (permanent fault over current, earth fault or any other type of selected and programmed event) the Auto-Recloser controller must send signals to the main server to inform control center about this fault.
- The supplier must supply one heavy duty laptop which can stand continuous operation.
- The software should be able to issue a number of privileges to verify that this user is authorized to communicate with Recloser through software security manager.
- Laptop Compatibility, A maintenance port, RS-232 or USB port should be provided on the front panel of the controller for local programming, troubleshooting and data analysis.



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- The Supplier should provide user-friendly communications software on a CD for use with a...EDC provided laptop computer. This software package should include help screens and an operator's manual. The software should be capable of operating under Windows XP®, Windows 7®, windows8 and windows10. Future upgrades to software should be made available at no charge & a serial connection will be installed with each control Panel.
- The programming and control software should enable graphical simulation for each line by drawing the single line diagram of each line, this will allow simulation of possible faults, coordination between upstream and downstream Reclosers with the main switchgear, and this will also allow calculation of possible short circuits current.
- The software should also be capable to calculate the recommended protection settings (Auto -calculate, manual coordination, manual calculation) of protection settings.
- This software should be able to control the Auto-Recloser and perform close and open operations for each Auto-Recloser remotely through control center.
- **SCADA Status/Control.**
 - 1- Local Status Indication. The controller should be capable of monitoring local status indication for SCADA purposes and should include as a minimum:
 - a. Switch Position (Open/Closed)
 - b. Battery Test (Bad/Good)
 - c. Local Mode (On/Off)
 - d. Remote/Auto Mode (On/Off)
 - 2- Required Analog Points. The following analog points should be available for SCADA.
 - a. Single-phase active power (Watts).
 - b. Single-phase reactive power (VARs).
 - c. Fault Indication (with the magnitude).
 - 3- SCADA Control Functions. The controller should have the capability of disabling the protection functions and reclose function, such that the



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Recloser will operate only as a remote switch via SCADA.

4- The controller should be capable of performing the following required remote functions:

Recloser Switch	Open / Close
Automatic (Recloser) Function	Enable / Disable

5- Fault event record memory: should be exist with 1000 event minimum and should be involved the faulty time, type of fault (over current or earth fault) and the operation sequence until the fault is cleared.

8. Additional Requirements:-

8.1 Marking

- i) The ratings and data of the Auto-Recloser should be provided in the nameplate which should be weather and corrosion resistant and indelibly engraved.**
- ii) The nameplate should be provided on the lower part of the tank and should consist of the following details:**
 - a) Manufacturer's name/trademark.**
 - b) Type designation/Serial number.**
 - d) Rated voltage and BIL.**
 - e) Rated continuous current and short time current/duration.**
 - f) Year of manufacture.**
 - g) Country of Origin.**

8.2 Routine Tests

The following routine tests As per ANSI/IEC should be carried out on all Auto-Reclosers; copy of the routine test report should be delivered with each Auto-Recloser.

- a) Reclosing and Overcurrent Trip Calibration.**
- b) Control, Secondary wiring and Accessory device check test.**
- c) Dielectric withstands test one minute dry power-frequency.**
- d) No load mechanical operation tests.**
- e) Partial discharge tests.**



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f) Main contacts resistance test.

g) Tightness tests.

8.3 type tests:

Complete type test report as per relevant ANSI / IEC C37.60.20^{١٢} standards must be submitted from independent west European, Canada, Kema or American laboratory.

- Dielectric tests on main circuits.
- interrupting current tests
- Partial discharge test.
- Temperature rise test.
- Mechanical operation test for 30,000 CO cycles.
- Degree of protection IP degree for control cubicle and Auto-Recloser tank contact resistance test of main circuits.
- Line charging and cable charging current test.
- Short-time withstand current and peak withstand current tests
- Critical current
- Making current tests
- Rated symmetrical interruption test (standard operation duty)
- Minimum tripping current tests
- Time-current tests
- X-radiation test for vacuum interrupters as per subclause 6.11 of IEEE C37-60-2012

9. Technical Literature and Drawings

Drawings and Manuals: The vendor should provide a copy of the manufacturer's latest drawings and manuals for each Auto- Recloser supplied. The drawings should include the general arrangements, dimensions, and weights of the reclosers. The manuals should include operating, programming and maintenance instructions. The Vendor should provide any installation information, performance or factory test data necessary in the field testing Auto-Recloser to determine normal operation.



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10. Warranty

10.1 Warranty.

- All equipment provided should be fully warranted by the manufacturer for defects in material and workmanship for a period of 18 months from delivery and 12 months from the date installation to the medium voltage network in Egypt before and the period will increase to 3 years (36 months) in the case of supply and installation for the first time.
- In addition, insulating materials should be warranted against deterioration and corrosion due to environmental exposure for a period of ten years from date of delivery.
- The manufacturer should carry all direct costs of correction and replacement of the defective materials or workmanship during the guarantee period, including transportation costs.

10.2 After sale service.

The tenderer should have a licensed service center in Egypt covers all its deliverables.... EDC reserves its right to claim a free after sale service all over the warranty period guaranteed by the tenderer that covers all its deliverables after sale service should include and not limited to the following:-

- Technical support.
- Maintenance.
- Original spare parts.
- Submittals.
- The supplying company undertakes training for the specialists in the distribution companies to enable maintenance and programming of the on site Auto-Recloser according to the requirements of the distribution company.



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- All the services, spare parts, materials and accessories will be needed during the after-sale maintenance should be guaranteed by the tenderer immediately and wherever required.

10.3 Factory Visits & Fabrication Inspection(option)

EDC Reserves its right to instruct the tenderer to arrange a priceless fabrication, inspection and a factory visit for 3 of its representatives or as per mutually agreed to ensure the tenderer/vendor compliance with the tender.

10.4 Training

EDC Reserves its right to claim a priceless training for (3) of its representatives on the vendor's factory, site, lab or as per mutually agreed.

The training should be include and not limited to the following:

- Testing & commissioning.
- Troubleshooting.
- Operating and maintenance.
- The supplying company undertakes training for the specialists in the distribution companies to enable maintenance and programming of the onsite Auto-Recloser according to the requirements of the distribution company.

10.5 Information to be supplied with the offer

The following should be furnished with the offer.

- a) Catalogues describing the equipment and indicating the model No.
- b) Literature describing the operational feature of the equipment.
- c) Constructional features, materials used for components and relevant technical literature.
- d) Overall dimensional drawing of the Auto-Recloser including general arrangement.
- e) Dimensional drawing of mounting arrangement.
- f) Quality Assurance Certification Conforming to ISO 9001, 14001.
- g) Completed schedule of guaranteed technical particulars (Annexure A)



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- h) Reference list for the Auto-Recloser in Egypt with at least 5 years reference list.
- i) Reference list for the Auto-Recloser worldwide proving that the manufacture has at least 10 years Auto-Recloser manufacturing experience.
- j) Test Certificates should be supplied along with the offer in soft copy (CD). The test certificates should be issued by a recognized independent testing authority in America, Canada or Europe.

Test Certificates and performance curves, based on the type test conforming to relevant standard should be supplied along with the offer. The type test certificates should clearly identify the equipment concerned, showing the manufacturer's identity, model & type and basic technical parameters. The test certificates referred to should be issued by a recognized independent testing authority acceptable to the Purchaser.

- يتم تقديم خطاب اعتماد من الشركة القابضة (ساري) والمشاريع المنفذة (سابقة الأعمال) في مصر لضمان استخدامه.
- يشترط في توريد أي معيد تيار جديد أن يتوافق مع النظام القائم والمعمول به (Software) في أي شركة توزيع يقبل الدخول علي نظام SCAD System والتزام الشركة الموردة بربط الأجهزة مع نظام التحكم القائم بشركة التوزيع .

Failure to furnish any of the above particulars will result in the offer being rejected.

11. Guarantee Table

The tenderer should be filled in all guarantee tables attached to these specifications accurately. The offer which will not be accompanied with a complete copy of drawings, catalogues and filling guarantee tables will not be considered.



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Annexure A
SCHEDULE OF GUARANTEED TECHNICAL PARTICULARS
(TO BE FILLED BY THE TENDERER)

Manufacturer information	unit	...EDC	Vendor Proposed Values
Name of manufacturer	---	Required	
Country of origin	---	Required	
Type of Auto-Recloser	---	Required	
Manufacturer years of experience (year)	---	Required	
Service Conditions			
operating temperature	°C	55	
Minimum ambient temperature	°C	- 5	
operating humidity	%	95	
operating solar radiation	KW/m ²	1.1	
operating attitude	meter	1000	
Basic operating parameters			
Recloser tank construction material	---	According to the specification	
Thickness of Recloser tank	mm	3	
Thickness of the control cubicle	mm	2.5	
Weight of tank <=100Kg	KG	100	
Bushing type	---	According to the specification	
Creepage distance of insulator	mm	4 cm/KV	
Type of interrupting method medium	---	Vacuum	
Type of insulation method material	---	Solid	
Minimum allowable duty cycle at Rated short-circuit breaking current	---	According to the specification	
Recovery time for 4 th Open	sec	60	



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Current sensing	---	According to the specification	
Number of current sensors per Recloser	NO	6	
Voltage sensing	---	According to the specification	
Number of voltage sensors per Recloser	NO	6	
control cable length	m	7	
Degree of protection for the Recloser tank	---	Sealed IP65	
Degree of protection for the control cubicle	---	Sealed IP65	
Expected life time of Recloser	years	25	
Maintenance interval	---	Maintenance free	

DESIGN REQUIREMENTS

Standard	---	IEEE C37.60, IEC 62271-111		
Nominal operating voltage (Max.)	KV	12	22	
Service voltage	KV	11	24	
Rated continuous current	A	630	630	
Rated power frequency	HZ	50	50	
Rated lightning impulse withstand voltage	KV	95	125	
Rated power-frequency withstand voltage, 1 min, dry	KV	50	60	
Rated power-frequency withstand voltage, 10 sec, wet	KV	45	50	
Rated partial discharge decay level, not less than (at 10 Pc).	KV	13.2	17.2	
Rated symmetrical making current peak	A	40	31.5	
Rated short-circuit breaking current	KA	16	12.5	
Rated short time current withstand (3sec)	KA	16	12.5	



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Cable charging current	A	10	25	
Line charging current	A	2	5	
Surge arrestors				
No of lightning arrestors	---	3 or (6 as option)		
Nominal current	KA	10		
Operating voltage	KV	12	24	
Max continuous rating voltage	KV	10KV±5%	20KV±5%	
Leakage path	Cm/KV	4 cm/KV at least		
Number of interrupting operations				
minimum number of electrical operations at rated current (630Ampere)	---	≥30,000 CO cycle		
minimum Interrupting life operations at rated breaking current	---	200 CO cycles		
Type of operating mechanism				
Closing	---	Magnetic actuator		
Tripping	---	Magnetic actuator & opening spring		
Main contact resistance	μΩ	≤60		
Closing time	m sec	55-80		
Opening time	m sec	15-45		
Breaking time	m sec	25		
control cable length	m	9		
Power supply cable	m	9		
Earthing cables	m	9		
Sensor accuracy				
Rogowski current sensor	±%	Not less than 1 %		
Capacitive voltage sensor	±%	Not less than 1 %		
Uninterruptible Power Supply				
Source of AC input voltage	V	According to the		



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		specification	
Type, rated burden and application standard of auxiliary supply voltage transformer	---	According to the specification	
Battery type	---	Sealed	
Battery voltage	V	12	
Battery capacity	Ah	≥ 25	
Battery operating time	HR	≥ 48	
Battery recharge time	HR	≤ 12	
Battery replacement interval	Year	≥ 10	
Modem or radio power supply voltage	V	5:15 , step 0.5	
Modem or radio power supply max. continuous power output	W	15	
Auto - Reclose			
Duty cycle	---	O-0.1s-CO-1s-CO-1s-CO-60s-C	
Number of sequential tripping operation to lockout.	NO	Adjustable up to 4	
First Auto-reclose dead time	Sec	0.1 – 180	
Second Auto-reclose dead time	Sec	1 – 180	
Third Auto-reclose dead time	Sec	1 – 180	
Reset time	Sec	1 – 180	
Timing resolution	Sec	0.01	
Event logging			
Number of the events	NO	1000	
Characteristics of event	----	According to the specification	
Integration period	Minute	5/10/15/30/60	
Number of registers	NO	9000 covering interval from 31.25.....375 days	
Measurements functions	----	According to the	



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		specification	
Protection functions	----	According to the specification	
Sensitive Earth Fault protection (SEF)			
Operational pick up current range	A	1 – 80	
Operational pick up current resolution	A	1	
SEF operating time resolution	A	0.01	
Cold Load Pick up			
Cold Load multiplier range	---	1 – 2	
Cold Load multiplier resolution	---	0.1	
Cold Load time	Minute	1 – 60	
Cold Load time resolution	Minute	1	
Inrush harmonics filter	---	Present , based on detection of second harmonic	
Zone Sequence Coordination	---	present	
under voltage element (UV)			
Pick up phase voltage range	P.U	0.6 Ur – Ur	
Pick up phase voltage resolution	P.U	0.01 Ur	
Tripping time range	sec	0.1 – 100	
Tripping time resolution	sec	0.01	
Current unbalance protection element (CU)			
Pickup setting range	P.U	0.6....1*I1	
Time setting range	Sec	0.1....100	
Bolted fault protection element (BF)			
Pickup current range(instantaneous)	A	20....6000	
Voltage unbalance protection element (VU)			



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Pickup setting range	P.U	0.05....1*U1	
Time setting range	Sec	0.1....100	

High current lockout (Live line protection element) LL

Pick up current	A	10 - 1280	
Current setting resolution	A	1	
Tripping time setting range	Sec	0...2	
Time resolution	Sec	0.01	

Under frequency protection element

Pickup frequency	HZ	45....50	
Pickup frequency resolution	HZ	0.01	
Tripping time	Sec	0.1 - 180	

Voltage Reclosing control

Voltage multiplier range	P.U	0.6 - 1 U rated	
Voltage multiplier resolution	P.U	0.01	

Life time counters

Mechanical wear	%	Required	
Contact wear	%	Required	
Total number of close and open operations.	NO	Required	

Fault counters

	--	According to the specification	
Fault profile recording	--		

UPS monitoring

	--	present	
Mechanical position indicator	--	present	

Manual tripping

	--	Manual tripping ring easily operable with a hook-stick.	
Lifting tackle	--	present	



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Drainage filter	--	present	
Anti-pumping duty	--	present	
Lockout ring	--	present	
RTU and Communication protocol	--	According to the specification IEC 60870-5- 104 or DNP	

I / We guarantee the technical data given above for the offered AUTO-RECLOSER.

Signature:

Date.....